Budget constraints and radical compression for Diophantine consistency

Research contribution prepared for Vladimir Reshetnikov

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Abstract

We combine a centered sign budget with intersective interpolation, parity, and five-root fusion to obtain fixed ordinary polynomials in eleven and twelve unrestricted integer unknowns. Every named coefficient is an explicit, authenticated integer; all other named registers are polynomial abbreviations, not quantified variables. A second reduction absorbs exact equations into a one-sided bounded positivity certificate. The constructions preserve solvability precisely through explicit witness maps and named ordinary native/Pell/refutation calibration theorems. Their unsolvability is therefore provably equivalent in ZFC to syntactic Con(ZFC) under those inherited results. This is neither a proof of consistency nor a new end-to-end kernel certificate. We state exact domains, degrees, operation conventions, dependencies, and reproducible finite checks; we make no global minimality or historical-priority claim.

1 Results and interpretation

The objective is a fixed ordinary polynomial, not an equation with variable exponents, a schematic proof predicate, or an uncompiled arithmetic circuit with hidden existential wires. We count only declared unknowns. Literal arithmetic circuits use binary addition, subtraction, and multiplication; squaring and multiplication by a fixed numeral each cost one operation. Sharing a previously computed register costs nothing. These are arithmetic circuit counts, not bit-operation costs.

The central new combination removes a guard coordinate and replaces large variable radical weights by the fixed coefficients $1, 2, 4, 8$. A rational projection first recovers the main root; an even shared strong ordinate makes the other four radicands odd, so their square-class contributions cannot cancel by 2-adic valuation. A second new argument uses an upper bound, rather than an absolute-value bound, on a positivity certificate to impose exact equalities without a separate finalizer.

The later power-synchronization constructions are deliberately presented as intermediate interfaces. Their exposed-coordinate and graph counts do not constitute a variable count for an ordinary polynomial. The complete polynomial results and those interfaces must not be conflated.

1.1 Complete polynomial endpoints

The following table separates the integer and positive domains. Degree and operation count are independently checked after specializing the actual fixed coefficients. These counts exclude no existential graph witnesses.

Domain and mechanism	Variables	Degree	Operations
Integers, interpolated parity, direct	11	2912	200
Integers, auxiliary center	11	2272	204
Integers, two centers	11	2240	206
Integers, centered budget and trace	12	1000	180
Integers, cheaper companion	12	1024	179
Integers, bounded-guard absorption	15	176	125
Strictly positive integers	10	912	126

The eleven-integer construction combines the late intersective interpolation lemma with the centered budget and a parity separator. The twelve-integer construction offers substantially lower degree and evaluation cost. Neither is a variable-count record against the repository's ten-integer constructions, which have much greater degree and use different reductions. The nine-positive and direct quartic families likewise remain distinct comparison points.

2 Pinned comparison and inherited interfaces

The comparison snapshot is the public `VladimirReshetnikov/ProveIt` main commit

`7e533c33819ee1f703b69f2778a767cdc069940d`,

observed on 24 October 2026 at 03:32 UTC. It is 32 commits ahead of `55ca592b2f42` (full identifiers are retained in the source ledger). We distinguish the repository's current artifacts from the immediately preceding centered compression packet, recovered separately with research-archive SHA-256

`f68edc30d832d880785c69e2277af1efa`
`1c9b7eca04289eb8b4e3fdf8b3b4885`.

The incoming archive bearing the additional word *Dependency* contains only the earlier Joint Guards archive; it is not the research packet.

2.1 Small-variable ordinary equations

Every operation count charges each live binary addition, subtraction and multiplication, including squaring and multiplication by a constant. Fixed numeral leaves and register reuse are free. Named registers are polynomial abbreviations rather than further quantified variables. These are measures of the supplied circuits, not lower bounds.

Domain	Variables	Degree	Operations	Source family
$\mathbb{Z}$	13	1088	195	Guard-first root fusion
$\mathbb{Z}$	13	1392	186	Guard-first root fusion
$\mathbb{Z}$	13	1408	185	Guard-first root fusion
$\mathbb{Z}$	14	652	163	Compact fixed-prime norm
$\mathbb{Z}$	14	664	156	Compact fixed-prime norm
$\mathbb{Z}$	14	696	150	Root fusion
$\mathbb{Z}$	14	748	148	Compact fixed-prime norm
$\mathbb{Z}$	14	792	137	Root fusion
$\mathbb{Z}$	15	204	128	Primitive parity norm
$\mathbb{Z}$	15	352	124	Parity norm, sum of squares
$\mathbb{Z}$	16	106	123	Previous centered compression
$\mathbb{Z}$	16	112	119	Modulus-four parity recovery
$\mathbb{Z}$	16	160	115	Parity norm, sum of squares
$\mathbb{Z}$	17	48	110	Previous centered compression
$\mathbb{Z}_{>0}$	10	1152	131	Four-root radical/residue
$\mathbb{Z}_{>0}$	11	211	111	Adaptive double centering

The first parity family's fifteen-variable degree-204 equation strictly improves the preceding fifteen-variable degree-300, 132-operation centered equation. Conversely, the previous sixteen-variable degree-106, 123-operation equation dominates the newly reviewed prime-separation point of degree 108 and 132 operations. The sixteen-variable degree-112, 119-operation parity equation is a genuine tradeoff with degree 106, while it dominates the previous degree-138, 122-operation sum-of-squares alternative. The previous seventeen-variable degree-48, 110-operation equation also dominates the root-fusion quotient companion of degree 88 at the same cost. The table is a local comparison, not a claim of global optimality or historical priority.

The exact complete polynomial packets are mapped in `SOURCE_MAP.md`. Useful proof sources are the first parity archive's `PROOF15.md`, the second parity archive's `MOD4_PROOF.md`, and the root-fusion report's `crt1088.tex`. The compact fixed-prime candidate preserves its entire parent polynomials and replaces generic four- and five-root norm evaluators by exact circuits with respectively 43 and 93 operations. These are pointwise identities, stronger than existence equivalence.

2.2 Combining the reductions

The parity-15 reduction uses radicands $A = 4a+1$, $B = 4b+1$ and a nonzero integer C, with weights $1, 2, 4$. In an irrational squareclass the least weighted coefficient has parity preventing cancellation. The primitive eight-sign norm is divisible coefficientwise by 256. Its application must prove $Q \neq 1$ before recovering the deleted guard and retain the exceptional $Q = 0$ branch. The modulus-four reduction instead recovers an omitted odd strong multiplier through a rational complementary root and a private-root sign choice. These are different mechanisms, not equivalent free-polynomial rewritings.

Root fusion first obtains realness of the guard root, then positivity and noncancellation, and only then rationality and integrality. Completeness can require fresh compatible canonical Pell witnesses. Fixed-prime separation uses signed valuations before positivity; one cannot silently import a positive-radicand hypothesis.

The small-degree integer families retain the whole source-34 tuple: $\beta = 2^{1064921} - 1$, already shifted M_F, and determinant scales 2^{1113} and 2^{666104}, with the remaining calibrated ports. The six literal payloads total 615883 raw bytes. Positive ten- and eleven-variable families instead use the distinct compact-44 tuple. Individual ports must not be exchanged between these bindings.

2.3 A direct route independent of K215 and Jones

The fixed-power-bound candidate gives complete degree-four polynomials in 3443 nonnegative variables, with 2838 quadratic rows, 15754 collected monomials and coefficient height 127302193, or in 5960 nonnegative variables with height 1992. The separate prime-lift signed-domain polynomial has 8619 unrestricted integer variables, 4619 quadratic rows and 39811 monomials. These are materialized ordinary polynomials, not merely uncompiled special-function interfaces.

The improvement fixes each positive-power certificate's common bound at $W = n + k$, with fresh canonical completeness witnesses, removing the bound and two slacks at each of 55 calls. Earlier shared-CRT and beta-majorant proofs remain essential. This route uses actual C222 and named ordinary ZF/ZFC-refutation interfaces, not K215 or Jones.

2.4 Verification boundary

The replay receipts distinguish source authentication, finite arithmetic identities, coefficient reconstruction, literal DAG rebuilding and unbounded mathematical arguments. None decides consistency or supplies a new end-to-end Lean or encoded internal-ZFC derivation. The formalization pause and inherited native/compiler/Pell/refutation interfaces are respected.

Root fusion is replayed with its complete release verifier, normally and under optimized Python. The first parity archive's complete verifier explicitly requires ordinary Python and rejects optimized execution by policy. Its seven separate algebraic auditors are nevertheless replayed under optimized Python. A policy rejection is not a successful optimized whole-release run. Exact invocations and outcomes are recorded in `replay/results/REPLAY_SUMMARY.json`.

3 Late arrivals and attribution

Before finalizing the comparison, we inspected three further incoming archives at main commit

`96563ce58ef39cb6681cf6cfcc17975f17c3dfaf`,

observed on 24 October 2026 at 03:50 UTC. This is a separately pinned supplement to the earlier snapshot, not an implicit change of source versions. The archives are *Twelve Integer Interpolation*, *Congruence Certificates*, and *Shared Pell 608*.

The trace argument already appears in a concurrent arrival. The interpolation archive's thirteen-variable companion explicitly uses normalized field trace before radical separation to recover the cubic Y scale. It removes both the v^2 guard multiplier and $\Lambda = 4H + 1$ main-root weight, obtaining degree 1072 with 190 operations. Accordingly, early rational projection and those two removals are not claimed here as new to the inspected repository. The exact source is that archive's `proofs/TRACE13_PROOF.md`.

The present twelve-variable route instead combines the earlier centered-budget guard with deletion of its nearest-square coordinate. The residue obstruction modulo 3 recovers the main root before separating the other four roots. That centered-budget reduction and its distinct completeness map are not supplied by the inspected late trace proof. This is a comparison of concrete packets, not a claim of worldwide priority.

The late twelve-variable construction is different. It retains the nearest-square coordinate and shares a positivity Pell witness with the strong witness through $I = src^2$. The monotone

intersective polynomial

$$\Pi(x) = (x^2 + 31)(x^3 + x + 1)$$

excludes the zero-multiplier branch and restores the main congruence. A full-residue Pell interpolation theorem chooses one strong pair (I, f) first and then realizes every auxiliary residue $T \bmod I$. The resulting complete twelve-variable polynomials have degree 4832 with 248 operations and degree 16000 with 223 operations. They are relevant baselines even though their mechanisms differ.

Domain	Variables	Degree	Operations	Late source
$\mathbb{Z}$	12	4832	248	Interpolation
$\mathbb{Z}$	12	16000	223	Interpolation
$\mathbb{Z}$	13	1072	190	Rational trace
$\mathbb{Z}$	14	414	148	Congruence, retained root
$\mathbb{Z}$	15	218	130	Congruence, retained roots
$\mathbb{Z}$	16	124	118	Congruence, retained roots
$\mathbb{Z}_{>0}$	12	70	98	Free-J positive form
$\mathbb{Z}_{>0}$	13	51	88	Free-J positive form

The fourteen-variable point dominates the earlier degree 652/163-operation, 664/156, 696/150 and 748/148 points; degree 792/137 remains a lower-operation alternative. The late fifteen-variable point is already dominated by the preceding parity degree-204/128-operation point. The sixteen-variable degree-124/118-operation point remains a tradeoff with degree 112/119 and 160/115. The positive improvements do not supply a new ten-positive-variable equation.

The Congruence archive also factors a repeated component of the first biquadratic squaring to save three multiplications. The present 83-operation cleared five-root evaluator improves further, but its comparison should acknowledge that intermediate exact simplification, rather than attributing every saved gate to a previously absent idea.

The direct quartic frontier also changed. The Shared Pell archive supplies a literal quartic in 608 nonnegative variables, with 491 quadratic residuals, 7582 collected monomials and coefficient height 79824571. Its unrestricted-integer companion has 1685 variables, 850 residuals, 12082 monomials and height 79827802. It shares a Pell environment across 25 positive-power calls and 20 distinct exponent certificates, uses one radix for 14 binomial graphs, and proves a self-bounding factorial representation. These are ordinary direct C222 polynomials, independent of K215 and Jones, and supersede the earlier 3443-natural/8619-integer variable counts in the inspected direct branch. Different coefficient-height tradeoffs remain relevant.

Additional verification scope. The three downloaded Git blobs and all 419 internal manifest entries were authenticated. An independent static census checked topology, liveness and operation totals for the eight supplied integer DAGs. Independent expansion of all 491 and 850 quadratic residuals agrees coefficient-for-coefficient with both Shared Pell quartics. No late upstream executable was run, and no full imported semantic calibration was re-established. Exact degree claims for the high-degree norm graphs are taken from their supplied ordinary proofs and degree certificates, not represented as fresh expanded checks. The records are `late/FETCH_LEDGER.json`, `late/STATIC_AUTHENTICATION.json`, `late/DIRECT_EXPANSION_CHECK.json` and `late/LATE_COMPARISON.md`.

4 Twelve integers by centered-budget root fusion

The preceding thirteen-integer CRT construction spends one coordinate on a nearest-square joint guard. We remove that coordinate by fusing the centered-budget guard with the five-root norm. The key is an early rational-squareclass projection: it recovers the cubic scale before individual radicals are known to be rational, then forces the main root to be integral through a fixed obstruction modulo 3. This also removes the old guard multiplier and the inflated main-root coefficient.

The preferred equation has **twelve unrestricted integer variables, exact total degree** 1000**, and** 180 **live binary arithmetic operations**. An auxiliary-only centering gives degree 1024 in 179 operations. The incoming CRT endpoint has thirteen integers, degree 1088, and 195 operations under the same counting convention. These are comparisons of explicit constructions, not optimality or historical-priority claims. The same six fixed source34 numerals are retained. A late-arriving trace/interpolation precursor already obtains thirteen integers, degree 1072, and 190 operations by trace before separation, removing the guard multiplier and enlarged main-root coefficient. We credit that parallel source explicitly; the present improvement combines its trace method with the centered-budget guard and the modulo-3 main-root projection. Its proof is retained in `provenance/trace_precursor/TRACE13_PROOF.md`.

4.1 The literal polynomial

Supply exactly the twelve independent integers

$$j, C, z, h, r, v, w, T, c, n, p_2, p_3 \in \mathbb{Z}.$$

Uppercase C and lowercase c are distinct. Every other symbol in this section is a polynomial abbreviation or proof notation, never a hidden existential variable. The fixed coefficients are $\beta, K_0, M_C, M_F, A_c, B_c$, where

$$\beta = 2^{1064921} - 1, \qquad A_c = 2^{1113}, \qquad B_c = 2^{666104}.$$

The other three native ports are the authenticated source34 values; M_F is already shifted and is not shifted again. Define

$$\begin{aligned}
J &= 6j + 1, \quad q = \beta J + 1, \quad W = 4(3w + 1), \quad X = qW, \\
Y &= qv, \quad E = XY, \quad a = (X + 1)Y, \quad H = 4a + 3, \\
\Delta &= a^2 + 4a + 3, \\
F &= (K_0 + W)C - z(q - 1), \quad \alpha = q - F - C, \\
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
k &= R + 1 + hE, \quad \eta = c - Yk, \quad i = 2r + 1, \\
D_f &= 1 + \Delta i^2 c^4, \quad Q = \Delta(D_f - 1), \quad \delta = Q - 1, \\
V &= D_f(cT - R) - c, \quad C_y = QV^2 - 1, \quad L = XY^2 k, \\
D_t &= 1 + L(L + k), \quad D_m = 1 + \Delta c^2, \quad b_0 = X + ac.
\end{aligned}$$

The packed expression for R agrees identically with the original one because $(q - 1)(q - F) + \alpha = q(q - F) - C$. The new guard is

$$\begin{aligned}
D_{\mathrm{err}} &= (A_c F - q)^2 + (B_c C - q)^2, \\
B &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\
A_g &= B - 1 - p_2^2 - p_3(p_3 + 1).
\end{aligned}$$

Finally put

$$M = 2q^2, \qquad U = MH, \qquad Z = U(n + V + L) + Mb_0 + H(v - q^2).$$

The term UL is an integral carrier shear. Omitting it gives the auxiliary-only companion.

Let $F_{32}(Z; A_1, \ldots, A_5)$ denote the product of $Z - \sum_{i=1}^{5} \epsilon_i \sqrt{A_i}$ over all $\epsilon \in \{-1, 1\}^5$. Invariance under formal sign changes makes this an integer polynomial, of degree at most 16 in each A_i. Our polynomial is

$$P_{12} = \delta^{16} F_{32}\left(Z; M^2 D_m, U^2 D_f, U^2 D_t, U^2 A_g, \frac{U^2 C_y}{\delta}\right). \tag{1}$$

The displayed denominator is completely cleared. One division-free interpretation, useful independently of the optimized circuit, works in the free rank-four algebra with $u^2 = d = M^2 D_m$ and $v_0^2 = e = U^2 A_g$. With $A = U^2 D_f$, $B_0 = U^2 D_t$, $C_0 = U^2 C_y$, and $\mathcal{R} = Z + u + v_0$, set

$$\begin{aligned}
\mathcal{J} &= \delta(\mathcal{R}^2 - A - B_0) - C_0, \\
\mathcal{K} &= \mathcal{J}^2 - 4(\delta^2 A B_0 + \delta C_0 (A + B_0)), \\
\mathcal{N} &= \mathcal{K}^2 - 64 \delta^3 A B_0 C_0 \mathcal{R}^2.
\end{aligned}$$

The product of the four formal conjugates of $\mathcal{N}$ is exactly (1). All operations are polynomial even when the specialized algebra is not a field. The optimized evaluator in `five_square_norm.py` computes this same polynomial; the norm appendix gives its 83-operation implementation and the complete 20,349-coefficient identity check.

4.2 The noncircular reality bootstrap

For every integer input, q is even, $q \equiv 2 \pmod 3$, and $|q| \geq 16$; also $|X| \geq 64$. If $v = 0$ then $a = 0$, $\Delta = 3$, $H = 3$. Otherwise $|Y| \geq 16$ and $|a| \geq 63 \cdot 16 = 1008$. Consequently Δ is always positive and odd, $H \neq 0$. Moreover $X \equiv 2 \pmod 3$, hence $a \equiv 0 \pmod 3$, $H \equiv 0 \pmod 3$, and $b_0 \equiv 2 \pmod 3$. Since $H \equiv 3 \pmod q$ and $q \equiv 2 \pmod 3$, $\gcd(H, 2q^2) = 1$ unconditionally.

The denominator δ never vanishes at an integer input. For $c = 0$, $Q = 0$, $\delta = -1$, and $C_y/\delta = 1$. For $c \neq 0$, $Q > 1$ and $D_f > |c| + 1$. Since $b = cT - R$ is integral, $V = D_f b - c$ has $|V| \geq 1$. Thus

$$\frac{C_y}{\delta} = V^2 + \frac{V^2 - 1}{Q - 1} \geq 1.$$

The other three non-guard radicands are positive as well: for D_t use $s(s + 1) \geq 0$ for the integer $s = XY^2$.

Suppose $P_{12} = 0$. A complex linear factor of its norm vanishes. If $A_g < 0$, its radical supplies the only nonzero imaginary part of that factor, which is impossible. Hence $A_g \geq 0$, and

$$B = A_g + 1 + p_2^2 + p_3(p_3 + 1) > 0.$$

It follows that $B > 0$, $k \neq 0$, and

$$D_{\mathrm{err}} < q^2, \qquad 4\eta^2 < k^2. \tag{2}$$

No Pell root, native sign, or calibration theorem was used to obtain these facts. In particular, $v \neq 0$ has not been assumed; it will follow from the early projection below.

4.3 Early integer projection and main-root recovery

We use the following elementary projection principle. Suppose a signed sum of square roots of nonnegative integers, with integer coefficients, plus an integer multiple of a nonnegative rational-square root y is rational. If every rational value of y allowed by the construction is integral, the sum is an integer. Indeed, decompose each radical into a rational multiple of a positive squarefree radical. Independence of distinct squarefree radicals identifies the sum with its squareclass-1 part. Every term in that part is integral; irrational terms can cancel but contribute nothing to it.

Here every rational $y = \sqrt{C_y/\delta}$ is integral even before any scale recovery. For $c = 0$ it equals 1. Otherwise $Q > 1$ globally and adjacent-coefficient Pell descent applies to $QV^2 - (Q-1)y^2 = 1$. To recall the proof, write $y = a/s$ in lowest terms; then $s^2 \mid Q-1$. The norm-preserving transformation

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

keeps V' integral and the reduced denominator s unchanged. With positive signs and $|V| > 1$, it gives $-|V| < V' < |V|$; the norm excludes $V' = 0$. Descent reaches $|V| = |y| = 1$, proving $s = 1$.

Write d, f, t, g, y for the nonnegative roots of $D_m, D_f, D_t, A_g, C_y/\delta$. A vanishing factor has independent signs and reads

$$Z = M\epsilon_d d + U(\epsilon_f f + \epsilon_t t + \epsilon_g g + \epsilon_y y).$$

Divide by M. Since $U/M = H$ is integral, the projection principle gives $Z/M \in \mathbb{Z}$. But

$$Z/M = H(n + V + L) + b_0 + H(v - q^2)/M.$$

Thus $M \mid H(v-q^2)$, and the unconditional coprimality gives

$$v_{\text{old}} = (v-q^2)/(2q^2) \in \mathbb{Z}, \qquad v = q^2(2v_{\text{old}} + 1), \qquad Y = (2v_{\text{old}} + 1)q^3.$$

In particular $v \neq 0$ and $|Y| \geq |q|^3 \geq 4096$. This inference does not assert that each individual radical is already rational.

Now project the undivided factor onto squareclass 1. Every projected f, t, g, y contribution is integral. If d were irrational, its projection would be zero, forcing $H \mid Z$. That is impossible: $H \equiv 0 \pmod 3$ whereas $Z \equiv Mb_0 \equiv 1 \pmod 3$. Thus d is rational and integral. Put $d_s = \epsilon_d d$. The same projection modulo H gives $H \mid M(d_s - b_0)$, so

$$\gamma = (d_s - b_0)/H \in \mathbb{Z}.$$

The actual signed main root and its congruence have now been restored without separating the remaining four radicals.

4.4 Four-root separation on all signed inputs

Set $x = |X|$, $m = |Y|$, $K = |k|$, and $C_* = |c|$. The budget and early CRT give

$$x \geq 64, \quad m \geq |q|^3 \geq 4096, \quad K \geq 1, \qquad K(m - \tfrac{1}{2}) < C_* < K(m + \tfrac{1}{2}).$$

In particular the integer C_* is at least 16. We do not assume $K \geq 2$ before first-root integrality. Direct estimates give

$$\begin{aligned} \sqrt{\Delta} &> (x-1)m - 3 > 3, \\ K(xm^2 - 1) < t &< K(xm^2 + 1), \quad t > 3C_*, \\ d &> ((x-1)m - 3)K(m-1) > \tfrac{3}{4}t. \end{aligned}$$

The last comparison follows from

$$4((x-1)m-3)(m-1)-3(xm^2+1)=m^2(x-4)-4m(x+2)+9>0:$$

already $m\ge 16$ gives $(x-4)m-4(x+2)\ge 12x-72>0$. Also

$$f^2-(C_*^2-1)d^2\ge(\Delta-1)C_*^2+2>0,$$

so $f>(C_*-1)d\ge 15d>2t$ and $f>3C_*^2$. The unmultiplied guard is particularly small:

$$g<|q|K,\qquad \frac{g}{C_*}<\frac{|q|}{m-1/2}\le\frac{|q|}{|q|^3-1/2}<\frac{1}{255}.$$

For the final inequality, use $|q|^2\ge 256$ and $|q|>1/2$.

If $b=cT-R=0$, then $V=-c$ and $Q>C_*^2>1$, so

$$C_*^2<y^2=C_*^2+\frac{C_*^2-1}{Q-1}<C_*^2+1.$$

Thus $y>g$ and $y+g<C_*+1+C_*/255<3C_*<t$. The order f,t,y,g is strictly superincreasing. If $b\neq 0$, then $|V|\ge f^2-C_*>2f$, hence $y>2f$ and the order y,f,t,g is strictly superincreasing. Zero g is omitted.

Subtract the already integral main-root term from the norm factor and divide by U. The remaining signed sum of f,t,g,y is rational. Independence of distinct squarefree radicals now makes every term rational: the largest term in any nontrivial square class cannot cancel the smaller ones. Integer radicands make f,t,g integral, and the same adjacent-coefficient descent makes y integral.

4.5 Exact source restoration

Let $f_s,t_s,p_{1,s},y_s$ be the signed integer roots in the original vanishing factor. The early projection has already recovered the exact source scale, signed main root d_s, and integral γ.

We map this zero pointwise to the normalized centered-budget seventeen-integer source, not to the old determinant-guard source. Set its radix coordinate to $J_3=2j$, take

$$f_{\text{source}}=f_s,\quad v_{\text{scale}}=v_{\text{old}},\quad w_3=w,\quad \tau_{\text{source}}=t_s-L,\quad y_{\text{source}}=y_s-V,$$

and retain h,r,T,c,z, together with the restored F,α,γ. The six source residuals vanish: the first and auxiliary rows are the exact integral translations of the recovered norms; the main row uses $d_s=X+ac+H\gamma$; the transport row vanishes identically; and the normalized strong row is $f_s^2=1+\Delta i^2c^4$. In particular the source's packed $V=f_s^2(cT-R)-c$ is exactly our V.

Finally, since $A_g=p_{1,s}^2$, the same coordinates already satisfy

$$B=1+p_{1,s}^2+p_2^2+p_3(p_3+1)>0.$$

This proves a genuine zero of the pinned centered-budget17 source. Its ordinary soundness theorem supplies a finite syntactic ZFC refutation. No implication from the disk budget to the old determinant guard is asserted; the source's late half-binomial argument is what restores its original positivity guards.

4.6 Canonical completeness and scope

Conversely, use the authentic canonical source34 family associated to a finite refutation. Its prescribed half-binomial Y, unchanged by normalized strong/auxiliary completion, satisfies

$$D_{\mathrm{err}} \le q^2/2, \qquad 0 < \eta/k < 1/4, \qquad B > q^2k^2/4 > 0.$$

The centered source proves these estimates uniformly for that specified Y; there is no appeal to choosing Y freely or taking a limit. The canonical repunit J is odd and is 1 modulo 3, so $j = (J-1)/6$ is integral. Every positive integer N has a representation $1 + p_1^2 + p_2^2 + p_3(p_3+1)$: apply the three-square theorem to $4N-3$, which is 1 or 5 modulo 8, and separate one odd and two even squares. Set $v = Y/q = q^2(2v_{\mathrm{old}}+1)$ and represent B as $1 + p_1^2 + p_2^2 + p_3(p_3+1)$. Thus $A_g = p_1^2$. With canonical integral roots $d = b_0 + H\gamma, f, t, y$, choose

$$n = \gamma + f + t + p_1 + y - V - L - \frac{v - q^2}{M}.$$

Every term is integral, and substitution makes the displayed norm factor vanish. This proves completeness.

The carrier translation $n \mapsto n + L$ is an integer-polynomial bijection for fixed remaining coordinates. It changes neither the existence theorem nor the CRT restorations. Its exact degree reduction is a separate leading-form calculation, supplied in `DEGREE_PROOF.md` and checked by `check_degree.py`; no uncancelled gate-degree estimate is presented as an exact degree. The two literal circuits are `budget12_tau_centered.json` and `budget12_aux_centered.json`.

Thus, under the same pinned ordinary native/compiler and calibration theorems, ZFC proves

$$\left[\forall \mathbf{x} \in \mathbb{Z}^{12},\ P_{12}(\mathbf{x}) \neq 0\right] \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

This is neither an assumption nor a proof of consistency. It adds an unbounded algebraic reduction, not a new native-kernel certificate or encoded object-language ZFC derivation. The local provenance directory contains the centered-budget17/18 proof interfaces and the root-fusion ordinary dependencies; `SOURCE_PINS.json` authenticates those copies. Exact proof-cut diagnostics check the restoration identities, positive-coefficient root-bound certificates, and agreement of all six coefficient payloads. Such finite checks complement, but do not replace, the unbounded argument and its explicitly inherited calibration.

4.7 A division-free norm and exact degree certificates

The five-root product used here is an explicitly specified integer polynomial, not a call to a square-root oracle. The following evaluator works over every commutative ring. Its inputs are x, a, b, c, d, C, δ; it equals

$$\delta^{16} \prod_{\epsilon \in \{-1,1\}^5} \left(x - \epsilon_1\sqrt{a} - \epsilon_2\sqrt{b} - \epsilon_3\sqrt{c} - \epsilon_4\sqrt{d} - \epsilon_5\sqrt{C/\delta} \right)$$

where the displayed interpretation uses $\delta \neq 0$, while the evaluator itself is a polynomial even at $\delta = 0$. Define

$$
\begin{aligned}
x_2 &= x^2, \quad s_0 = c + d, \quad t_0 = a + b, \quad J_0 = \delta(x_2 + s_0 - t_0) - C,\\
K_0' &= J_0^2 + 4\delta\{\delta(s_0 x_2 + cd - ab) - Ct_0\},\\
K_1' &= 4\delta x(J_0 + 2\delta d),\\
K_2' &= 4\delta x(J_0 + 2\delta c),\\
K_3' &= 4\delta(J_0 + 2\delta x_2), \quad Z_0 = 64\delta^3 abC,\\
f_0 &= (K_0')^2 + c(K_1')^2 + d(K_2')^2 + cd(K_3')^2 - Z_0(x_2 + s_0),\\
f_1 &= K_0'K_1' + dK_2'K_3' - Z_0 x,\\
f_2 &= K_0'K_2' + cK_1'K_3' - Z_0 x,\\
f_3 &= K_0'K_3' + K_1'K_2' - Z_0,\\
g_0 &= f_0^2 + 4\{df_2^2 - c(f_1^2 + df_3^2)\},\\
g_1 &= f_0 f_2 - 2cf_1 f_3.
\end{aligned}
$$

The answer is $g_0^2 - 16dg_1^2$. The primes on the K_i' distinguish these local registers from the fixed native numeral K_0.

For verification, work in the free rank-four algebra with basis $1, u, v, uv$, $u^2 = c$, $v^2 = d$. The cleared three-root norm at $x + u + v$ has coefficients $f_0, 2f_1, 2f_2, 2f_3$. Taking its norm first under $u \mapsto -u$ and then under $v \mapsto -v$ gives exactly the evaluator. This is a free-algebra identity and makes no independence assumption about the square classes. The supplied literal DAG shares the repeated products and uses 83 binary operations. The inspected predecessor has 93; a late concurrent packet already supplies an intermediate factoring improvement, which we credit rather than treating all factoring as new. An independent reverse-order conjugate expansion agrees on all 20,349 nonzero monomials.

For the preferred twelve-variable specialization, exact sparse expansion at named polynomial cuts gives a degree upper bound of 1000, including all leading cancellations caused by the auxiliary and first-root centers. A nonzero actual-coefficient certificate is

$$[j^{456} w^{152} v^{168} h^{16} r^{64} c^{128} z^{16}] P = 2^{952} 3^{608} \beta^{456} > 0.$$

The exponents sum to 1000. Thus the certificate proves the exact degree after specializing all six fixed ports, rather than merely computing a formal degree before specialization. The cheaper companion has exact degree 1024, with

$$[j^{464} w^{160} v^{176} h^{16} r^{64} c^{128} z^{16}] P' = 2^{960} 3^{624} \beta^{464} > 0.$$

The files `guard/DEGREE_PROOF.md` and `guard/check_degree.py` contain the complete cut expansion, weights, strict component comparisons, and coefficient calculations. The independent audit also evaluates the literal circuits on a different actual-coefficient univariate line, obtaining a nonzero coefficient at each claimed top degree. This second check is a transcription diagnostic; the general upper bound and exact nonzero monomial are the degree proof.

5 Eleven integers by intersective interpolation and parity

The centered-budget guard can also replace the nearest-square guard in the late twelve-integer intersective construction. A shared Pell coordinate, an early main-root projection, and four odd

radicands then remove all variable separating weights. The result is an ordinary polynomial in **eleven unrestricted integers, of exact degree** 2240**, evaluated in** 206 **live binary operations**. Its less-centered companions have degree 2272 in 204 operations and degree 2912 in 200 operations. The late intersective predecessor has twelve integers, degree 4832, and 248 operations. The twelve-integer degree-1000 construction remains a lower-degree, lower-operation alternative.

The intersective quintic, full-residue interpolation, and trace-before- separation have explicit prior sources in the supplied provenance; no novelty for those individual devices is claimed. The new composition uses the centered budget, an even shared strong ordinate, fixed 2-adic separation, and integral norm centers. All comparisons concern these concrete equations, not global minimality or a new formal kernel certificate.

5.1 Complete definition

Supply exactly

$$j, C, z, h, r, v, w, T, c, n, s \in \mathbb{Z}.$$

Use the same six authenticated source34/cone numerals $\beta, K_0, M_C, M_F, A_c, B_c$, with $\beta = 2^{1064921} - 1$, $A_c = 2^{1113}$, $B_c = 2^{666104}$, and M_F already shifted. All remaining symbols are polynomial abbreviations, not additional coordinates:

$$\begin{aligned}
J &= 6j + 1, \quad q = \beta J + 1, \quad W = 12w + 4, \quad X = qW, \\
Y &= (2v + 1)q^3, \quad E = XY, \quad a = (X + 1)Y, \quad H = 4a + 3, \\
\Delta &= a^2 + H, \quad F = (K_0 + W)C - z(q - 1), \quad \alpha = q - F - C, \\
R &= (q(q - F) - C)(q^2 - 1) + (M_C + qM_F)J, \\
k &= R + 1 + hE, \quad \eta = c - Yk, \quad L = XY^2k, \\
D_{\mathrm{err}} &= (A_cF - q)^2 + (B_cC - q)^2, \\
\mathcal{P} &= k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2, \\
D_g &= 1 + 4(3\mathcal{P} - 4)s^2, \quad I = 2src^2, \\
D_f &= 1 + \Delta I^2, \quad Q = \Delta^2 I^2, \quad \delta = Q - 1, \\
V &= D_f(c - R + 4IT) - c, \quad C_y = QV^2 - 1, \quad D_a = \delta C_y, \\
D_t &= 1 + L(L + k), \quad D_m = 1 + \Delta c^2, \quad b_0 = X + ac.
\end{aligned}$$

The subscript a in D_a denotes the auxiliary radical, not a new variable. Set $u = HT - b_0$ and define the two quadratic-algebra coefficients

$$\begin{aligned}
A_\pi &= u^5 + 10u^3D_m + 5uD_m^2 + 32H^2(u^3 + 3uD_m) \\
&\qquad + H^3(u^2 + D_m) + 31H^4u + 31H^5, \\
B_\pi &= 5u^4 + 10u^2D_m + D_m^2 + 32H^2(3u^2 + D_m) + 2H^3u + 31H^4.
\end{aligned}$$

In the formal algebra $d^2 = D_m$ they satisfy

$$A_\pi + B_\pi d = ((u + d)^2 + 31H^2)((u + d)^3 + H^2(u + d) + H^3).$$

Finally put

$$S = IH^5, \qquad \Theta = (2Q - 1)V + 4L, \qquad Z = S(n + \Theta) - A_\pi.$$

For the integer polynomial F_{32} defined by its 32 formal signed linear factors, the equation is

$$F_{32}(Z; B_\pi^2 D_m,\, 4S^2 D_a,\, S^2 D_f,\, 16S^2 D_t,\, 64S^2 D_g) = 0. \tag{3}$$

Here every radicand is an integer polynomial and there is no denominator to clear. The literal circuit uses only binary addition, subtraction, and multiplication. The direct and auxiliary-only companions replace Θ by 0 and $(2Q-1)V$, respectively. These are invertible integer-polynomial translations in n, not additional constraints.

5.2 The intersective polynomial and the zero-scale exclusion

The prior interpolation construction uses

$$\pi(x) = (x^2+31)(x^3+x+1) = x^5+32x^3+x^2+31x+31.$$

It has a root modulo every nonzero integer, no zero in a real extension of $\mathbb{Q}$ of degree at most two, and positive derivative on $\mathbb{R}$. For completeness, at an odd prime other than 31, either the cubic has a root or its irreducible Frobenius 3-cycle makes its discriminant -31 a square. These roots are simple. At 31, the cubic has simple root 3; at 2, $-31 \equiv 1 \pmod 8$ is a 2-adic square. Prime-power lifting and CRT give intersectivity. The quadratic has no real root and the cubic has no rational root, so neither factor has a zero in a real field of degree at most two. Finally

$$\pi'(x) = 5x^4+96x^2+2x+31 > 0.$$

For $H \neq 0$, write $\Phi_H(x) = H^5\pi(x/H)$. Then $A_\pi \pm B_\pi\sqrt{D_m} = \Phi_H(u \pm \sqrt{D_m})$ are nonzero. Moreover $B_\pi > 0$, either by monotonicity or by the exact certificate

$$B_\pi = 5u^4+10u^2D_m+D_m^2+32H^2D_m+95H^2u^2+H^2(u+H)^2+30H^4 > 0.$$

For every integer input, q is even, $|q| \geq 16$, $|X| \geq 64$, $|Y| \geq 4096$, and $|a| \geq 63 \cdot 4096$. Thus $H \neq 0$, $\Delta > 1$ is odd, and D_m, D_f, D_t are positive. Also $Q = (\Delta I)^2 \neq 1$, so $\delta \neq 0$. If $I = 0$, $D_a = 1$. If $I \neq 0$, then $c \neq 0$, $|I| \geq c^2$ and $D_f > |c|+1$. For the integer $b = c - R + 4IT$, $V = D_f b - c$ has $|V| \geq 1$. Therefore $C_y > 0$ and $D_a > 0$.

At a zero of (3), $S = 0$ would specialize the norm to $(A_\pi^2 - B_\pi^2 D_m)^{16} \neq 0$. Hence $S \neq 0$, and I, s, r, c are nonzero. If $D_g < 0$, a vanishing complex factor would have precisely one nonzero imaginary summand, with coefficient $8S$. This is impossible. Thus $D_g \geq 0$, which for nonzero integer s forces $\mathcal{P} \geq 2$. In particular

$$D_{\mathrm{err}} < q^2, \qquad |\eta/k| < \tfrac{1}{2}.$$

The disk inequality implies $0 < A_c F/q < 2$ and $0 < B_c C/q < 2$. Since $A_c, B_c \geq 8$, F, C, α all have the same nonzero sign as q. Also

$$|k|(|Y| - \tfrac{1}{2}) < |c| < |k|(|Y| + \tfrac{1}{2}), \qquad \operatorname{sign}(c) = \operatorname{sign}(kY).$$

Positivity of η has not been presumed.

5.3 Early main-root projection and parity extraction

A vanishing factor can be written, with independently signed roots, as

$$S(n+\Theta) = A_\pi + B_\pi d + S(2\widetilde{y} + f + 4\tau + 8g).$$

All five underlying radicands are nonnegative integers. Project this identity onto the rational squareclass. Every rational radical is an integer, while irrational squareclass terms have zero rational projection. If $d = \pm\sqrt{D_m}$ were irrational, it would follow that $S \mid A_\pi$, hence $H \mid A_\pi$.

But $q \equiv 2$, $W \equiv 1$, and $X \equiv 2 \pmod 3$, so

$$a \equiv H \equiv \Delta \equiv 0, \qquad b_0 \equiv 2, \quad u \equiv D_m \equiv 1 \pmod 3.$$

Consequently $A_\pi \equiv 1 + 1 + 2 \equiv 1 \pmod 3$, contradicting $H \mid A_\pi$. Thus the actual signed main root d is integral. Projecting again yields

$$IH^5 \mid A_\pi + B_\pi d = \Phi_H(d - b_0 + HT).$$

Monicity forces $H \mid d - b_0$: if $p^e \| H$ and the argument had valuation $b < e$, its leading fifth power would have valuation $5b$, strictly lower than every other term, contradicting divisibility by p^{5e}. Hence $\gamma = (d - b_0)/H$ is integral.

Now subtract the main-root contribution and divide by S. The signed sum $f + 2\widetilde{y} + 4\tau + 8g$ is rational. All four radicands are odd:

$$D_f \equiv D_a \equiv D_g \equiv 1 \pmod 4, \qquad D_t \equiv 1 \pmod 8.$$

Indeed I is even, $Q \equiv 0 \pmod 4$ and $\delta \equiv C_y \equiv -1 \pmod 4$; X is a multiple of 8. A positive odd integer radicand is an odd square times an odd squarefree integer. Within any nontrivial squareclass, the participating signed coefficients therefore have distinct 2-adic valuations from $0, 1, 2, 3$. The unique least valuation cannot cancel. Independence of distinct squarefree radicals implies that all four roots are rational, and hence integers. No comparison of their magnitudes is required.

The rational $y = \widetilde{y}/\delta$ satisfies $QV^2 - (Q-1)y^2 = 1$ and is integral by adjacent-coefficient descent. Explicitly, a reduced denominator b has $b^2 \mid Q - 1$; the norm-preserving transformation

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

keeps V' integral and the denominator unchanged, while reducing $|V|$ to 1. The terminal value has $|y| = 1$, proving integrality.

5.4 Soundness with an even strong multiplier

The recovered equations and the exact definitions give

$$f^2 - \Delta I^2 = 1, \qquad \rho^2 = \Delta(f^2 - 1), \qquad \rho^2 V^2 - (\rho^2 - 1)y^2 = 1, \qquad \rho = \Delta I,$$
$$d = X + ac + H\gamma, \quad d^2 - \Delta c^2 = 1, \quad \tau^2 - XY^2(XY^2 + 1)k^2 = 1,$$
$$V = fo - c = cj_{\text{aux}} - R, \qquad k = R + 1 + hE,$$

where

$$o = f(c - R + 4IT), \qquad j_{\text{aux}} = 4\Delta s^2 r^2 c^3 (c - R) + 8srcD_f T.$$

The incoming multiplier is even. We do not assert a nonintegral restoration such as $(2\Delta sr - 1)/2$.

Here is the exact modification of the centered source's signed rank proof. Its common-sign packing bounds give, for $Q_0 = |q|$,

$$Q_0(Q_0^2 - 1) < R < Q_0^4 - 2, \qquad |E| > 3R + 1.$$

Put $B_0 = |a + 2|$ and $P_0 = |2XY^2 + 1|$. First/main Pell classification, using only the already proved centered interval and sign identity, gives $|k| = 2\psi_{P_0}(n_0)$, $|c| = \psi_{B_0}(p)$, $p > n_0 \geq 24$, and $|c| > 2p, 2R$. The first-index congruence has positive branch $n_0 \geq (R + 1)/2$ and negative branch $n_0 > R$.

The strong norm gives $|I| = \psi_{B_0}(m)$ and $|f| = \chi_{B_0}(m)$. Since $c^2 \mid I$, strong divisibility first implies $p \mid m$. Write $m = pt$ and $D = \chi_{B_0}(p)$. Composition and binomial expansion give

$$|I| = |c|\psi_D(t), \quad D^2 - 1 = \Delta c^2, \quad \psi_D(t) \equiv tD^{t-1} \pmod{|c|}.$$

Since D is coprime to c, $|c| \mid t$. Thus $p|c| \mid m$ and $m > 2p$. These are precisely the strong-rank inputs; oddness was not used.

For the auxiliary parameter $|\rho| > 1$, its Pell index ℓ is odd because $\chi_{|\rho|}(2t) \equiv (-1)^t \pmod{|\rho|}$, while its abscissa is divisible by $|\rho|$. The exact odd-index quotient identities, together with $V \equiv -c \pmod{|f|}$ and $V \equiv -R \pmod{|c|}$, yield

$$\psi_{B_0}(\ell) \equiv \pm\psi_{B_0}(p) \pmod{|f|}, \qquad \ell \equiv \pm R \pmod{|c|}.$$

The signed abscissa step-down then gives $\ell \equiv \pm p \pmod{m}$. Since $|c| \mid m$ and $0 < p, R < |c|/2$, $p = R$. The negative first-index branch is impossible; no-wrap gives $2n_0 = R + 1$. The unchanged sign-uniform ratio and main-root congruence prove $|Y| > 2^R$ and $X = \mathrm{sign}(c)2^R$. Since $X \equiv 2 \pmod 3$ and R is odd, $X = 2^R$, $c > 0$; the surviving first-index sign then gives $k, Y > 0$.

The centered source's uniform half-binomial estimate now applies. Writing $R = 2t + 1$, it gives an even integer M_{bin} with

$$\frac{M_{\mathrm{bin}}}{2} < \frac{c}{k} < \frac{M_{\mathrm{bin}}}{2} + \frac{1}{4}.$$

Its proof bounds the reciprocal binomial tail by $1/8$ and the Pell ratio error by $12t/(2^R + 1) < 1/8$, uniformly for the prescribed Y. Since $|c/k - Y| < 1/2$ and Y is integral, $Y = M_{\mathrm{bin}}/2$ and $0 < \eta/k < 1/4$. Thus $\eta, k - \eta > 0$.

An exact old odd-coordinate chart can now be restored with fresh private witnesses, rather than by changing the parity of incoming I. Let $A_0 = a + 2$ and put $I_* = \psi_{A_0}(Rc)$, $f_* = \chi_{A_0}(Rc)$. Here A_0 is even, R, c are odd, and composition gives $c^2 \mid I_*$; therefore I_* and $\rho_* = \Delta I_*$ are odd. If $t = (R - 1)/2$ and $Q_t(z) = \chi_{\sqrt{z}}(2t + 1)/\sqrt{z}$ is the odd-index quotient polynomial, choose

$$V_* = (-1)^{t+1}Q_t(\rho_*^2), \qquad y_* = \psi_{\rho_*}(R).$$

Its two evaluations give $V_* \equiv -c \pmod{f_*^2}$ and $V_* \equiv -R \pmod c$. Hence the old strong and auxiliary coordinates are integral, with odd multiplier ρ_*/c^2, and preserve $k = R + 1 + hE$. The recovered positive guards and $X = 2^R$ restore the remaining old rows. The pinned ordinary native/compiler and finite-refutation theorem therefore applies. This regeneration works for every odd R on soundness, not only $R \equiv 3 \pmod 4$.

5.5 Full-residue interpolation and canonical completeness

We use the prior full-residue lemma in its precise order: for even $A_0 > 1$, $R \equiv 3 \pmod 4$, $c = \psi_{A_0}(R) \equiv R \pmod 4$, and any prescribed $N > 0$, one can first fix $I, f > 0$ with

$$f^2 - (A_0^2 - 1)I^2 = 1, \qquad 4Nc^2 \mid I,$$

such that *every* residue class modulo this same I contains a T for which

$$V = f^2(c - R + 4IT) - c, \qquad ((A_0^2 - 1)I)^2V^2 - (((A_0^2 - 1)I)^2 - 1)y^2 = 1$$

has an integer solution.

For precision, finite order of the Pell recurrence modulo $4Nc^2$ selects a positive even index m_0 and such a fixed I, f. The correct period modulo f^2 is $\lambda = 4m_0f$, not merely $4m_0$. Indeed, with $\alpha = A_0 + \sqrt{A_0^2 - 1}$, $\alpha^{4m_0} \equiv 1 - 4f\alpha^{m_0} \pmod{f^2}$, whose fth power is 1. Lucas lifting gives $v_p(I) \geq v_p(m_0)$ for $p \mid I$; at 2 the exact formula is $v_2(I) = v_2(m_0) + v_2(A_0)$. Thus $\gcd(\lambda, 4I^2) \mid 4I$.

For a requested class $T \equiv a \pmod{I}$, the congruences

$$\ell \equiv R \pmod{\lambda}, \qquad \ell \equiv R - 4Ia \pmod{4I^2}$$

are compatible. Choose a positive solution, which is 3 modulo 4. For $S_0 = (A_0^2 - 1)I$, put $V_0 = \chi_{S_0}(\ell)/S_0$ and $y = \psi_{S_0}(\ell)$. The odd quotient identities give $V_0 \equiv -c \pmod{f^2}$ and $V_0 \equiv -\ell \pmod{4S_0^2}$, including the important factor 4. Consequently

$$T = \frac{V_0 + c - f^2(c - R)}{4If^2}$$

is integral and is a modulo I, since $f^2 \equiv 1 \pmod{I}$. This proves the simultaneous fixed-strong-witness interpolation needed below; its full valuation proof is retained locally with the precursor.

Start now with the authentic canonical source34 family. Its prescribed half-binomial Y gives $D_{\mathrm{err}} \leq q^2/2$ and $0 < \eta/k < 1/4$, so $\mathcal{P} > q^2k^2/4 \geq 64$. Put $D = 3\mathcal{P} - 4 > 0$. Since $D \equiv 2 \pmod{3}$, it is nonsquare. Square any positive Pell unit $g_0 + u_0\sqrt{D}$ to obtain

$$g = g_0^2 + Du_0^2, \qquad s = g_0u_0 > 0, \qquad g^2 - 4Ds^2 = 1.$$

Apply the interpolation lemma with $N = 2s$; it fixes I, f with $8sc^2 \mid I$. Set $r = I/(2sc^2)$. Choose an intersective root a modulo I of $\pi(a + \gamma)$; the already fixed I, f then supply $T \equiv a \pmod{I}$ and the auxiliary root y. With $\widetilde{y} = \delta y$, all five radicands are integer squares, and

$$A_\pi + B_\pi d = H^5\pi(T + \gamma)$$

is divisible by S. Finally set

$$n = \frac{A_\pi + B_\pi d}{S} + 2\widetilde{y} + f + 4\tau + 8g - \Theta.$$

This integer makes a norm factor vanish. The witness order is native data and budget, then s, then fixed I, f, r, then the intersective class and T, y, then n. Nothing chosen later changes the positive budget or reselects the strong witness after the desired class is known.

5.6 Exact metrics and the inherited conclusion

Center in the n carrier	Integers	Exact degree	Binary operations
0	11	2912	200
$(2Q - 1)V$	11	2272	204
$(2Q - 1)V + 4L$	11	2240	206

The degree certificates are independent of the existence proof. The exact identities behind the centers include

$$4D_a - ((2Q-1)V)^2 = -V^2 - 4Q + 4, \qquad 16D_t - (4L)^2 = 16Lk + 16.$$

The supplied leading-form calculation gives degree 2240 for the primary equation. On the actual fixed-numeral slice it has the nonzero coefficient

$$[j^{1176}w^{280}v^{296}h^{16}s^{120}r^{112}c^{224}T^{16}]\,P_{11} = 2^{2592}3^{1460}\beta^{1176} > 0.$$

The exponent sum is 2240. The exact sparse degree cut agrees with the formal root filtration $16 \cdot 91 + 8 \cdot 51 + 8 \cdot 47 = 2240$; the supplied coefficient proves that this upper bound is attained. The literal circuits and independent checkers report every live binary operation, with no uncounted division or root.

Under the same explicitly inherited ordinary native/compiler and calibration theorems, ZFC proves

$$[\forall \mathbf{x} \in \mathbb{Z}^{11},\ P_{11}(\mathbf{x}) \neq 0] \quad \Longleftrightarrow \quad \mathrm{Con(ZFC)}.$$

The result neither assumes nor proves consistency. The unbounded reduction, the inherited calibration, finite circuit and coefficient checks, and the separate proof-assistant/object-ZFC obligations remain distinct. The complete local source pins and retained notices identify which parts of the interpolation, centered-budget, and signed-Pell arguments are inherited.

6 Absorbing equations into a bounded positivity certificate

The second new construction has **fifteen unrestricted integer variables, exact degree** 176**, and** 125 **live binary operations**. The incoming parity equation has degree 204 and 128 operations. The improvement does not assert that its radical norm is nonnegative. Instead, a low-degree *upper bound* on a positivity certificate lets us absorb the remaining equations inside one radicand. The norm itself then becomes the final polynomial, without a unit product or outer square.

Lemma 6.1 (Bounded-certificate absorption)**.** *Suppose B, H are real-valued expressions with $H \geq 0$ and $B \leq \lambda H - \varepsilon$, where $0 \leq \lambda < 1$ and $\varepsilon > 0$. For integer residuals $r_1, \ldots, r_m$, put $S = \sum_i r_i^2$. Then*

$$B - HS \geq 0 \quad \Longrightarrow \quad H > 0 \text{ and } r_1 = \cdots = r_m = 0.$$

When the residuals vanish, replacing B by $B - HS$ changes nothing.

Proof. If $H = 0$, the proposed nonnegative expression is at most $-\varepsilon$. If $S \neq 0$, integrality gives $S \geq 1$, whence $B - HS \leq (\lambda - S)H - \varepsilon < 0$. Both are contradictions. □

6.1 Complete equation and the new substitution

The coefficient tuple is the same fixed source34 tuple as above; write $A_c = \texttt{cone_A}$ and $B_c = \texttt{cone_B}$. This construction uses a different affine chart from the twelve-variable one. Its independent variables are exactly

$$J, C, z, f, h, r, v, w, \gamma, T, c, s, p_1, p_2, t \in \mathbb{Z}.$$

Here s denotes the old `pair_root`, and t denotes `t_parity`. Define the ordinary polynomial registers

$$\begin{aligned}
J_0 &= 3J+1, \quad q = \beta J_0 + 1, \quad W = 4(3w+1), \quad X = qW,\\
Y &= (2v+1)q^3, \quad E = XY, \quad a = E+Y, \quad H_0 = 4a+3,\\
\Delta &= a^2 + H_0, \quad F = (K_0+W)C - z(q-1),\\
R &= (q(q-F)-C)(q^2-1) + (M_C + qM_F)J_0,\\
k &= R + 1 + hE, \quad \eta = c - kY, \quad \zeta = k - \eta,\\
D &= X + ac + H_0\gamma, \quad I = (2r+1)c^2, \quad V = fcT - f^2R - c,\\
G &= A_cFB_cC - (q - A_cF - B_cC)^2, \quad Z_0 = \eta\zeta,\\
B_0 &= GZ_0 - (s^2 - G - Z_0 + 1)^2 - 1 - p_1^2 - p_2^2.
\end{aligned}$$

The two formerly separate residuals are

$$r_m = D^2 - \Delta c^2 - 1, \qquad r_s = I^2 - \Delta(f^2-1).$$

The entire new substitution is

$$S = r_m^2 + r_s^2, \qquad B_* = B_0 - q^2k^2S, \qquad B_g = 4B_* + 1. \tag{4}$$

Previously $B_g = 4B_0 + 1$ and the two residuals were imposed separately. Finish the equation by setting

$$\begin{aligned}
A_4 &= q(3w+1)Y^2(XY^2+1)k^2, \quad A = 4A_4 + 1,\\
Q &= I^2, \quad \delta = Q - 1, \quad C_a = \delta(QV^2 - 1),\\
x &= 2t+1, \quad U = t(t+1) - A_4 - B_g + 4C_a,\\
M &= U^2 + 4x^2C_a - AB_g.
\end{aligned}$$

The requested single, division-free integer polynomial is

$$\boxed{P_{15} = M^2 - 16x^2U^2C_a.} \tag{5}$$

All displayed registers are abbreviations, not additional unknowns. The literal ordered specification is `finalizers/absorbed15_degree176.json`; the accompanying `absorbed15_degree176_dag.json` contains only binary addition, subtraction and multiplication gates.

6.2 The universal upper bound, including every sign case

Let $u = A_cF$ and $v_0 = B_cC$, solely for the following identities:

$$4(q^2 - 3G) = \big(3(u+v_0) - 4q\big)^2 + 3(u - v_0)^2, \qquad 4Z_0 = k^2 - (2\eta - k)^2.$$

Thus $G \le q^2/3$ and $Z_0 \le k^2/4$ without using any residual. If both are nonnegative, their product is at most $q^2k^2/12$. If $GZ_0 \le 0$, subtracting the square in B_0 leaves a nonpositive quantity before its additional $-1 - p_1^2 - p_2^2$. The remaining case has $G = -g$ and $Z_0 = -z_0$ with $g, z_0 > 0$; then

$$(s^2 - G - Z_0 + 1)^2 \ge (g + z_0 + 1)^2 > gz_0 = GZ_0.$$

Consequently the following is a genuinely universal upper bound:

$$B_0 \le \frac{q^2k^2}{12} - 1. \tag{6}$$

No positivity of G, Z_0, or the old guard was assumed. If B_g is an integer square, its residue 1 modulo 4 makes $B_g \ge 1$, so $B_* \ge 0$. Lemma 6.1, with $H = q^2k^2$, $\lambda = 1/12$ and $\varepsilon = 1$, now forces $r_m = r_s = 0$. This also handles $qk = 0$: that branch cannot give a square B_g.

6.3 Parity extraction and noncircular restoration

The unchanged three-root identity is

$$256P_{15} = \prod_{\epsilon_1,\epsilon_2,\epsilon_3\in\{-1,1\}} \left(x + \epsilon_1\sqrt{A} + 2\epsilon_2\sqrt{B_g} + 4\epsilon_3\sqrt{C_a}\right). \tag{7}$$

It is a polynomial identity, valid for arbitrary integer radicands, including negative values and $C_a = 0$. At a zero, a signed radical sum is rational. Write each nonzero radicand as an integer square times a signed squarefree integer and group by rational squareclass. Distinct classes are linearly independent over $\mathbb{Q}$. Since A and B_g are odd, their square factors are odd. A nontrivial class containing the A term has an odd coefficient plus even coefficients, so cannot cancel. A class containing B_g but not A has twice an odd coefficient plus a multiple of four, so cannot cancel either. A class containing only a nonzero C_a term also cannot cancel. Thus A, B_g are positive integer squares and C_a is a nonnegative integer square. If $C_a = 0$, simply omit its zero radical; the same argument still forces A and B_g to be squares.

The deduction order matters. First squarehood of B_g and (6) force *both* old residuals to vanish. Only now use $r_s = 0$ to establish $\delta \neq 0$. Indeed the affine scales give $|q| \geq \beta - 1 \geq 16$, $|X| \geq 4|q|$, $|Y| \geq |q|^3$, and hence $\Delta > 1$. The equation $I^2 = \Delta(f^2 - 1)$ excludes $I^2 = 1$. There is therefore no assumption of a nonzero auxiliary radicand hidden in the earlier squareclass extraction.

Choose integral roots $u^2 = A$, $d^2 = B_g$, and $e^2 = C_a$. For $Q > 1$, the adjacent-coefficient Pell denominator lemma makes $y = e/\delta$ integral from $QV^2 - (Q-1)y^2 = 1$; its elementary descent is retained in the source. For $Q = 0$, $C_a = 1$ and $\delta = -1$, so integrality is immediate. Put $P_0 = k + E(c - \eta)$ and restore the three deleted private coordinates by

$$\tau_{\text{old}} = 2u - 2P_0 + k, \qquad y_{\text{gap}} = y - V, \qquad p_3 = (d-1)/2.$$

These are integers and make every residual of the pinned seventeen-integer parent vanish. In particular $S = 0$ gives $B_* = B_0$, so the restored positivity equation is exactly the original one. Conversely, any parent zero has $S = 0$ and leaves the guard unchanged; its integral roots u, d, e give $t = (u + 2d + 4e - 1)/2 \in \mathbb{Z}$ and make (7) vanish. This is a two-way existence reduction using every parent zero, not just canonical witnesses.

6.4 Exact measurements and proof boundary

The residuals r_m, r_s have degree 14, while q^2k^2 has degree 16. Thus the added penalty has degree 44, exactly the old guard radicand's degree. Structural degree propagation bounds (5) by 176. At the authentic fixed coefficients, an explicit line has degree-176 coefficient 179645231 modulo 1000000007, which is nonzero. An independent finite-difference calculation agrees; a separately chosen independent line gives 215929316 modulo the same prime. These are lower-bound certificates paired with the universal structural upper bound.

The literal circuit has 24 additions, 29 subtractions and 72 multiplications: 125 operations in total, including squaring and constant multiplication. Independent checks verify all 15 variable leaves, all six fixed ports, liveness, local polynomial cuts, and the zero-radicand corner. Finite diagnostics supplement the proof, not replace it.

The construction imports the incoming parity package's explicitly retained native/Pell/refutation calibration and its original coefficient binding. Under that ordinary theorem, ZFC proves that integer nonsolvability of (5) is equivalent to syntactic Con(ZFC). No consistency proof, new end-to-end kernel certificate, or global minimality claim follows from these arithmetic and circuit checks.

7 One-sided residue control in ten positive variables

We obtain a fixed polynomial in ten strictly positive integer unknowns, of exact degree 912 and with 126 live binary arithmetic operations. The inspected four-root residue predecessor has degree 1152 and 131 operations. Thus the degree falls by 20.83%, with five fewer operations. The coefficient tuple and calibration are unchanged. This is not a nine-variable result or a claim of global optimality.

The useful refinement is to bound a rational main-carrier residue only *from above*. Its large negative values are harmless for the positive transport gap. An auxiliary translation and two root centers then allow a degree-two weight instead of the predecessor's degree-38 majorant. The full local proof and retained interfaces are in `positive/ten_centered/PROOF.md`; the argument below makes the new inequalities and choice order explicit.

7.1 Complete ordinary polynomial

The fixed ports are the coherent compact44 numerals β, K_0, M_C, M_F, with $\beta = 2^{590069} - 1$ and M_F already shifted. Their actual bytes and generation recipe are delivered. The only unknowns are

$$j, F, \alpha, i, T, X, s, n, \eta, \zeta \in \mathbb{Z}_{>0}.$$

Every subsequent symbol is a polynomial abbreviation:

$$\begin{aligned}
&J = 3j + 1, \quad q = \beta J + 1, \quad h = F + \alpha, \quad C = q - h,\\
&Y = q^3 s, \quad E = XY, \quad k = \eta + \zeta, \quad c = kY + \eta,\\
&a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \quad \ell = EkY,\\
&R = \big((q-1)(q-F) + \alpha\big)(q^2 - 1) + (M_C + qM_F)J,\\
&B_t = (qK_0 + X)C - qF, \quad I = ic^2, \quad D_f = 1 + \Delta I^2,\\
&Q = \Delta^2 I^2, \quad \delta = Q - 1, \quad T_o = T + h, \quad C_f = (a+2)I,\\
&V = D_f(cT_o - R) - c, \quad D_t = \ell(\ell + k) + 1, \quad D_d = 1 + \Delta c^2,\\
&L = kT_o, \quad W = 1 + (q-1)L, \quad v = k - R - 1,\\
&U = q^5, \quad S = q^4, \quad b_0 = X + ac, \quad M = (q-1)U.
\end{aligned}$$

Set

$$\begin{aligned}
Z = M\big(&H(\ell + k + n + L + s + V + C_f) + b_0\big)\\
&- HUWB_t + H(q-1)(SX - v),\\
A = M^2H^2D_t, &\quad B = M^2H^2D_f,\\
D = M^2D_d, &\quad C_N = M^2H^2(QV^2 - 1).
\end{aligned}$$

The final division-free kernel is

$$\begin{aligned}
&\sigma = A + B + D, \quad \pi = AB + (A+B)D, \quad \rho = ABD,\\
&a_N = \delta(Z^2 - \sigma) + C_N, \quad t_N = \delta C_N Z^2,\\
&k_N = a_N^2 + 4(t_N - \delta^2\pi), \quad r_N = 16\delta^3\rho,\\
&b_N = a_N k_N - r_N,\\
&e_N = k_N^2 + 16t_N a_N^2 - 4r_N(\delta Z^2 + C_N),\\
&\boxed{P_{10}} \ \boxed{= e_N^2 - 64t_N b_N^2 = 0.}
\end{aligned}$$

The graph uses 75 multiplications, 34 additions, and 17 subtractions; squares and scalar products are charged. Numeral leaves and reuse are free.

In proof only write $D_y = (QV^2 - 1)/\delta$. The exact kernel is δ^8 times the product of the sixteen factors

$$Z - \varepsilon_1 MH\sqrt{D_t} - \varepsilon_2 MH\sqrt{D_f} - \varepsilon_3 MH\sqrt{D_y} - \varepsilon_4 M\sqrt{D_d}.$$

The independent checker verifies the whole 495-term identity. All clearing factors are positive on the declared domain.

7.2 Root recovery precedes the sign bootstrap

Write t, f, y, d for the positive roots of D_t, D_f, D_y, D_d. The predecessor's unconditional separation proof remains valid with T_o in place of T. Indeed $q \geq 16$, $Y \geq q^3$, $c \geq 2Y + 1$, $a \geq 2Y$, and

$$\ell < t < \ell + k < (a+1)c, \qquad f > (a+1)c^2, \qquad 0 < d/H < c, \qquad t > 2c + 1.$$

If the integer $cT_o - R$ is zero, then $c < y < c + 1$ and the four roots are superincreasing in order $d/H, y, t, f$. Otherwise $y > |V| \geq D_f - c > f + t + d/H$, and their superincreasing order is $d/H, t, f, y$. Thus every nonempty signed subset sum is nonzero. Square-class character independence in a multiquadratic field shows that a rational signed sum forces each root rational. At a norm zero we therefore have signed roots τ_*, f_*, y_*, d_* and

$$\ell + k + n + L + s + V + C_f - \frac{WB_t}{q-1} + \frac{b_0}{H} + \frac{SX - v}{U} = \tau_* + f_* + y_* + \frac{d_*}{H}. \tag{8}$$

The first, strong and main roots are integral because their squares are integers. The adjacent-coefficient Pell descent makes y_* integral from $QV^2 - (Q-1)y_*^2 = 1$. It uses only integer $Q > 1$ and V: the transformation

$$(V, y) \longmapsto ((2Q-1)V - 2(Q-1)y, (2Q-1)y - 2QV)$$

preserves the reduced denominator of rational y, has integral first coordinate, and strictly decreases its positive absolute value until $|V| = 1$, where $|y| = 1$. No native bound or main-root orientation is used.

Integer first and main roots give Pell indices $m, p \geq 1$ with

$$|\tau_*| = \chi_{2XY^2+1}(m), \quad k = 2\psi_{2XY^2+1}(m), \qquad |d_*| = \chi_{a+2}(p), \quad c = \psi_{a+2}(p).$$

Here $p \geq 2$; $m = 1$ would force $k = 2$, $c = 2Y + 1 < 2(a+2)$, a contradiction. Thus, *before* $C > 0$,

$$k > 8XY^2, \qquad c > 8XY^3 \geq 8q^9. \tag{9}$$

The actual mask inequalities give the unconditional estimate $|R| < \widehat{R} = q^3(q+h)$. Since $h \geq 2$, (9) implies $ch > \widehat{R}$, and therefore

$$0 < cT_o - R < c(T + 2h), \qquad V > 1.$$

As $\delta - (\Delta - 1)D_f = \Delta(I^2 - 1) > 0$ and $\Delta - 1 > 4(Y + 1)$, the auxiliary gap satisfies

$$0 < y - V < \frac{V}{\delta} < \frac{c(T+2h)}{\Delta - 1} < \frac{k(T+2h)}{4} < \frac{L}{2}. \tag{10}$$

Moreover $f < C_f$, since $C_f^2 - D_f = I^2 - 1 > 0$.

Define rational proof coordinates

$$r = \tau_* - \ell - k - s + \frac{WB_t}{q-1}, \qquad \gamma = \frac{d_* - b_0}{H} - \frac{SX - v}{U}.$$

For either main-root sign, $d_* \le d < (a+2)c$ and $2c/H < k/2$, giving

$$\gamma < \frac{k}{2} + \frac{k}{q^5} + \frac{1}{q} + \frac{h}{q^2} < \frac{3k}{4} + \frac{h}{4} < k + \frac{h}{2} \le \frac{k(T+h)}{2} = \frac{L}{2}. \tag{11}$$

The middle bound uses $q \ge 16$, $h \ge 2$; the last uses $(k-1)(h-1) \ge 1$ and $T \ge 1$. There is no lower bound on γ. Equation (8) now reads $r + \gamma + f_* + y_* = n + L + V + C_f$. Equations (10)–(11) therefore give $r > n > 0$ for all root signs. Hence

$$\frac{WB_t}{q-1} = \ell + k + s + r - \tau_* > 0,$$

so $B_t > 0$, $C > 0$, and only now $3q < R < q^4 - 2$ is available.

7.3 Residues, reconstruction, and canonical completeness

The residue argument of the four-root predecessor applies unchanged: all added centers are integers. The normalized strong and signed auxiliary lemmas give $p = R$. Integrality of $r + \gamma$, together with $\gcd(H, q) = 1$, yields $q^5 \mid q^4 X - v$. The first Pell recurrence and $1 \le m < R < q^4$ then force $R = 2m - 1$ and $q \mid X$; consequently

$$(q-1)(d_* - b_0)/H \in \mathbb{Z}.$$

This step does not cancel H against $q - 1$. Instead Pell growth gives $Y > q2^R$. Modulo H, the main Pell polynomials satisfy

$$\chi_{a+2}(R) - ac \equiv 2^R, \qquad 2^R(\chi_{a+2}(R) + ac) \equiv 1.$$

A negative d_* would make $(q-1)(1+2^R X)/H$ an integer strictly between zero and one. For positive d_*, $(q-1)(2^R - X)/H$ is an integer with absolute value less than one. Thus $d_* > 0$, $X = 2^R$, and $H \mid d_* - b_0$. Now γ, r are integers and $(q-1) \mid B_t$. The normalized source is restored by

$$w = X/q, \quad h_{\mathrm{src}} = v/(XY), \quad \gamma_{\mathrm{old}} = (d_* - ac - X)/H,$$

$$f_{\mathrm{old}} = |f_*|, \quad \tau_{\mathrm{old}} = |\tau_*|, \quad y_{\mathrm{old}} = |y_*|, \quad T_{\mathrm{src}} = f_{\mathrm{old}} T_o, \quad z = 1 + \frac{(K_0 + w)C - F}{q-1}.$$

These are positive integers and satisfy every retained source equation.

Conversely choose canonical outer source witnesses and then a normalized strong pair $f, I = ic^2$. With $M_a = \Delta I$, the canonical auxiliary construction is

$$V^* = \chi_{M_a}(R)/M_a, \qquad y^* = \psi_{M_a}(R), \qquad T_o = \frac{V^* + c + Rf^2}{cf^2}.$$

For $R \ge 7$, its growth gives $T_o > M_a^4/c \ge c^7 > q > h$, and also $T_o > (a+2)I = C_f$. Thus $T = T_o - h > 0$. Form L, W only after these choices; then $L > C_f, k$. The canonical index gives $v \ge 4XY^2$, so

$$\gamma = \gamma_{\mathrm{old}} - X/q + v/q^5 > 0.$$

With $t_0 = B_t/(q-1) \ge q$, set $r = Wt_0 + \tau - \ell - k - s > qW - 2k > 2L$. Finally

$$n = r + \gamma + f + y^* - L - V^* - C_f > r - L - C_f > 0.$$

This gives (8) with positive roots and hence $P_{10} = 0$. No additional congruence on Y or auxiliary quotient is imposed. The choice order is outer source, strong pair, auxiliary pair, L, W, then r, n; in particular no cofinality or freshness assumption is hidden.

7.4 Exact degree and evidence boundary

Let $K = MH$, and scale the ten unknowns by a formal parameter λ. In the Laurent-series field over the rational-function coefficient field, choose roots with the indicated leading centers. Exact defects give

$$\begin{aligned} K\sqrt{D_t} &= K\ell + O(\lambda^{12}), \\ K\sqrt{D_f} &= KC_f - \frac{KI^2}{2C_f} + \text{terms of degree less than } 17, \\ K\sqrt{D_y} &= KV + O(\lambda^7), \\ M\sqrt{D_d} &= Mb_0 + O(\lambda^{11}), \\ Z &= K(\ell + C_f + V) + Mb_0 + O(\lambda^{15}). \end{aligned}$$

Here the large centers have degrees $21, 27, 49, 16$, respectively; the remaining strong-root defect has degree 17. Of the sixteen norm factors, eight have degree 49, four degree 27, two degree 21, and two degree 17. Their leading terms are nonzero. Clearing δ^8, of degree 336, gives

$$\deg P_{10} = 8 \cdot 49 + 4 \cdot 27 + 2 \cdot 21 + 2 \cdot 17 + 336 = 912.$$

Writing $q_0 = 3\beta j$, $k_0 = \eta + \zeta$, exact simplification yields

$$\mathrm{in}_{912}(P_{10}) = 2^{44} q_0^{558} i^{38} k_0^{86} X^{68} s^{154} (T + F + \alpha)^8.$$

Thus the coefficient of $j^{558} i^{38} \eta^{86} X^{68} s^{154} T^8$ is $2^{44}(3\beta)^{558} > 0$. The exact norm and circuit identities, six actual-coefficient finite-field degree replays, and ordinary and optimized Python runs are included. They certify formulas and measurements, not unbounded semantics. The semantic conclusion composes the proof above with the retained ordinary native/compiler calibration:

$$\mathrm{ZFC} \vdash (P_{10} \text{ has no strictly positive integer zero}) \longleftrightarrow \mathrm{Con(ZFC)}.$$

No consistency assumption or new end-to-end formalization is claimed.

8 Power fusion in the alternative tiling certificate

This section gives an arithmetic improvement, not a new numerical bound for ordinary Diophantine equations. For every fixed finite strict-boundary Wang tile set, the previous five-power interface can be replaced by either seven exposed natural coordinates and three power graphs, or eight exposed natural coordinates and two power graphs. All graph witnesses remain to be compiled and counted. The construction does not require Jones's universal formula, K215, or restriction to square boards.

8.1 Recovering geometry instead of guessing its exponents

Fix $k \geq 1$ tiles with edge colors (w_t, e_t, n_t, s_t). Color zero is required exactly on the outer boundary. Choose a power of two $C = 2^c$ greater than k and all colors, put $A_0 = 4C^2$, and choose a power of two $D \geq 32kA_0$. Initially use natural coordinates a, Q, G, Z, r and the graph

$$P = 1 + (C - 1)G = 2^{c(a+1)}.$$

Thus $P \geq C$ and $G \geq 1$. Set

$$H = A_0 P, \qquad R = DP^3, \qquad B = R/2, \qquad L = R^{2k}.$$

For a formal vector $x = (x_0, \dots, x_{k-1})$, write $U = \sum_t x_t$; W_0, E_0, N_0, S_0 denote its sums over tiles having the indicated edge color zero. The seven residuals $\ell_j(x) = \sum_t a_{jt}x_t + c_j$ are

$$\begin{aligned}
&U - G, \quad (Q-1)W_0 - (P-1), \quad QW_0 - CE_0,\\
&C\sum_t e_t x_t - \sum_t w_t x_t, \quad Q\sum_t s_t x_t - \sum_t n_t x_t,\\
&Q - 1 - (C-1)N_0, \quad QS_0 - PN_0.
\end{aligned}$$

Here x is not a tuple of certificate variables. It is recovered from Z. The third and sixth residuals have been reversed relative to the predecessor so that unbounded positive Q cannot create negative packed summands. Define ordinary polynomial expressions

$$\begin{aligned}
K_j &= \sum_{t=0}^{k-1}(H + a_{jt})R^{k-1-t},\\
F_j &= K_j Z + (B - HG + c_j)R^{k-1},\\
T_7 &= F_7 + Q\frac{L}{2P},\\
M_0 &= G\sum_{t=0}^{k-1} R^t,\\
M_1 &= R^{2k-1} - 1 - (R - 1 - B)R^{k-1},\\
M_* &= L - 1 - (R - 1 - B)R^{k-1},\\
A &= M_0 + M_1\sum_{j=1}^{6} L^j + M_* L^7,\\
Y &= Z + \sum_{j=1}^{6} F_j L^j + T_7 L^7.
\end{aligned}$$

The apparent quotient $L/(2P)$ means the ordinary polynomial $(D^{2k}/2)P^{6k-1}$. Impose the total-binomial graph $\binom{A}{Y} = 2r + 1$, with value zero when $Y > A$.

Bootstrap. Before bounding Q, every $H + a_{jt}$ is positive, every offset is at least $B - HP - P > 0$, and $F_7 \geq Z$. Binary Lucas parity gives $Y \subseteq_{\text{bits}} A < L^8$. Therefore

$$T_7 < L, \qquad Z < L, \qquad Q < 2P.$$

The bottom L-block now gives $Z \subseteq_{\text{bits}} M_0$, so $Z = \sum_{t<k} x_t R^t$ with $x_t \subseteq_{\text{bits}} G$ and $x_t < P$. This establishes the bounds before using any no-carry argument.

After these bounds, $|a_{jt}| \leq H/2$ and $|c_j| \leq 2P$. Noncentral convolution coefficients are less than $2kHG < R/8$. The central coefficient is

$$d_j = B + \ell_j(x) + H(U - G), \qquad 0 < B - HG - 2P \leq d_j \leq B + 2kHG + 2P < R.$$

Thus $F_j < R^{2k-1}$ with no carries. The guard occupies only the unused highest digit, because $QL/(2P) = (QR/(2P))R^{2k-1}$ and $QR/(2P) < R$. All lanes separate. Their central masks allow only the bit B, so $d_j = B$. The first row gives $(H + 1)(U - G) = 0$, and then every $\ell_j = 0$.

Geometry. Write $P = C^N$. The mask $G = 1 + C + \cdots + C^{N-1}$ and $k < C$ make $U = G$ an exact one-tile-per-cell identity. Boundary rows imply

$$(Q-1)W_0 = P - 1 > 0, \quad (C-1)N_0 = Q - 1, \quad QS_0 = PN_0.$$

Hence $Q \geq 2$, $\gcd(Q, N_0) = 1$, and $Q \mid P$. Since P is a power of two, $Q = 2^u$; $C - 1 \mid Q - 1$ gives $c \mid u$, so $Q = C^w$. Also $Q - 1 \mid P - 1$ gives $w \mid N$. These assertions use the elementary identity $b^s - 1 \mid b^t - 1$ if and only if $s \mid t$ for positive s, t. Taking $h = N/w$ recovers the rectangle, including its exact four boundary masks. The two color residuals check all adjacent edges. Conversely, a rectangle gives these masks and $a = wh - 1$; its guard is at most $L/2$, while $F_7 < R^{2k-1}$, so it satisfies the packed condition. This proves the five-coordinate, one-power/one-binomial interface for arbitrary rectangles.

8.2 A three-power interface with seven coordinates

Replace G by $g + 1$, so $P = C + (C-1)g \geq C$, and keep the same circuit. Use the seven coordinates Q, g, Z, v, q, s, t and require

$$\begin{aligned} Pv &= 2^{A+1}, \\ s + t + 1 &= 2^{(A+1)Y}, \\ (s + t + 1)(2q + 1) + s &= (Pv + 1)^A. \end{aligned} \tag{T3}$$

The first graph makes P a power of two and therefore a power of C. For $d = 2^{A+1}$, all binomial coefficients of $(d+1)^A$ are smaller than d. The last two equations say that its quotient by d^Y is odd, with a legitimate remainder $s < d^Y$. Thus they are exactly the binomial parity condition, including rejection of $Y > A$.

For completeness, a genuine tiling has $Y \leq A$, $P \leq A$, and $A \geq 10$. These bounds follow directly from the highest digit of M_* and $R \geq 512P^3$. Thus $2^{A+1}/P$ is an integer; choose it as v and use the Euclidean quotient and remainder. Geometry's separate power graph and area index have both disappeared.

8.3 Exact synchronization: two graphs and eight coordinates

The following elementary lemma removes another power graph. Suppose n, Y, d, e are positive, $d > n + 1$, $1 \leq Y \leq n$, $e = 2^n$, $d \mid e$, and

$$e - 1 = (d-1)\big(Y + (d-1)j\big).$$

Then $d = 2^u$ and $d-1 \mid e-1$ give $e = d^h$ with $h \leq n$. The displayed equation gives $h \equiv Y \pmod{d-1}$, so the bound forces $h = Y$. Without the size bound the conclusion is false: $e = 8, d = 2, Y = 1$ satisfies the congruence with $j = 6$.

Take the eight natural coordinates Q, g, Z, v, q, u, t, j and define

$$\begin{aligned} n &= (A+1)Y, & d &= P(n + v + 1), \\ e &= d(u + t + 1), & W &= e(2q+1) + du + 1, \\ S &= e - 1 - (d-1)\big(Y + (d-1)j\big). \end{aligned}$$

Require just the two graphs and one ordinary residual

$$e = 2^n, \qquad W = (d+1)^A, \qquad S = 0. \tag{T2}$$

Pre-bootstrap positivity gives $Y \geq 1$; $P \geq 2$ gives $d \geq 2(n+1) > n+1$. The lemma yields $e = d^Y$, whence $d = 2^{A+1}$. Since $P \mid d$, the required geometry follows. Moreover $0 \leq du + 1 < e$, as $e - (du+1) = d(t+1) - 1 \geq 1$. The second graph therefore has odd Euclidean quotient, exactly the binomial parity test.

For the converse set $d = 2^{A+1}$, $e = d^Y$, and write $(d+1)^A = em + s$ with $0 \leq s < e$. Here m is positive odd and $s \equiv 1 \pmod{d}$. Choose

$$q = \frac{m-1}{2}, \quad u = \frac{s-1}{d}, \quad t = \frac{e}{d} - u - 1, \quad j = \frac{(e-1)/(d-1) - Y}{d-1}, \quad v = \frac{d}{P} - n - 1.$$

All are nonnegative integers. For the only delicate bound, use $Y \leq A$, $P \leq A$, $A \geq 10$, and

$$2^{A+1} > A^3 + A^2 + A \geq P\big((A+1)Y + 1\big).$$

The exponential inequality holds at $A = 10$ and propagates by induction. This proves the eight-coordinate, two-power interface.

Honest ordinary-polynomial accounting. If exact fixed-base and general-base graph polynomials have respectively p_2, p_g internal natural witnesses, and total binomial has b, the three interfaces have respectively

$$5 + p_2 + b, \qquad 7 + 2p_2 + p_g, \qquad 8 + p_2 + p_g$$

ordinary-polynomial variables after sum-of-squares combination. In the last case the literal template is $E_2(n,e)^2 + E_g(d+1, A, W)^2 + S^2 = 0$, with independent internal witness tuples. Since $\deg A, \deg Y \leq 48k$, its degree is at most

$$2\max\{(96k+2)\delta_2, (96k+3)\delta_g, 192k+3\}.$$

The missing graph implementations, direct ZFC-checker tile instance and internal ZFC interpretation remain explicit obligations. The supplied standard-library checks and fixed-power expression DAGs validate the arithmetic implementation but are not a proof-assistant formalization.

9 Fixed numeral definitions and reproducibility

The main eleven-, twelve-, and fifteen-integer equations use one complete source34 tuple. Put $\rho = 2^{121}$ and $B = \rho^{8801} = 2^{1064921}$. The supplied file `guard/coefficients/recipe/wang943_tiles.json` is the entire fixed ordered list of 943 tiles, in west–north–east–south order. It is reconstructed, entry for entry, from all 430 transitions of the included fixed blank machine. The first two selectors are the specified start and forbidden tiles; their order is part of the numeral definition. No search for a halting computation enters this finite reconstruction.

For tile i let x_{it} be its four fixed labels. Set

$$\begin{aligned} W_j &= 2^{10+2j}, \qquad (o_0, o_1, o_2, o_3) = (1020, 1000, 980, 960), \\ E &= \{0, \ldots, 942\} \cup \{o_t + 2j : 0 \leq t < 4,\ 0 \leq j < 10\} \\ &\quad \cup \{1063, 1064, 1095, 1096, 2195, 3296\}, \\ c_i &= 1 + \sum_{t=0}^{3} \sum_{j=0}^{9} \mathrm{bit}_j(x_{it} + 512) W_{10t+j} + \sum_{j=40}^{43} W_j \quad (0 \leq i \leq 942). \end{aligned}$$

At a copy position put $c_{o_t+2j} = W_{10t+j}$; at the four anchors $1095, 1096, 2195, 3296$ put $W_{40}, W_{41}, W_{42}, W_{43}$ respectively. The two dummy coefficients at $1063, 1064$ are zero. Thus $|E| = 989$. Define the fixed integers

$$\begin{aligned}
\mu &= 1022 + \sum_{j=0}^{43} W_j, \\
Z_p &= (2^{119} - 1) \sum_{e=8792}^{8798} \rho^e + (2^{27} - 1)\rho^{8799}, \\
D_C &= \rho^{40} + \rho^{41} + \rho^{1100} + \rho^{2200} + \sum_{e \in E} c_e(\rho^{4393-e} + \rho^{5495-e}), \\
M_C &= B - 1 - \sum_{e \in E,\ e \neq 1} \rho^e - 2\rho^{1063}, \\
M_{F,n} &= 4 + \mu(\rho^{4393} + \rho^{5495}) + \sum_{j=0}^{9} (\rho^{1000+2j} + \rho^{1021+2j}) \\
&\quad + \rho^{2195} + \rho^{3296} + Z_p.
\end{aligned}$$

The actual polynomial ports are

$$\beta = B - 1, \quad K_0 = D_C + B\rho, \quad M_C, \quad M_F = M_{F,n} + B - 1,$$

with the two additional cone numerals $A_c = 2^{1113}$ and $B_c = 2^{666104}$. The six complete payloads total 615,883 uncompressed bytes. A payload with little-endian bytes $b_0, \ldots, b_{m-1}$ means precisely the integer $\sum_{i=0}^{m-1} b_i 256^i$, not a digest standing in for an unspecified number. Payload hashes, finite recipe reconstruction, complete tile equality, and the ordinary coefficient-adapter proof are all supplied.

The ten-positive construction deliberately uses a different complete compact44 tuple, with $\beta = 2^{590069} - 1$. Its four whole payloads total 295,388 bytes; its own source-specific recipe, transition list, adapter, and reconstruction check are retained under `positive/ten_centered/coefficients`. None of its ports is substituted into a source34 integer construction. In both tuples M_F has already been shifted; shifting it a second time would define the wrong polynomial.

10 Verification and mathematical boundary

The new unbounded reductions are proved in the preceding sections. Finite checks authenticate the fixed numerals, exact polynomial identities, cancellation-sensitive degrees, literal circuit transcription, and declared variable and operation counts. Finite tests cannot prove the unbounded native compiler or the refutation-calibration theorem.

The inherited results are retained with their source identifiers and ordinary proof statements. An independently kernel-checked composite theorem for these new literal equations, or an explicit encoded object-language ZFC derivation, is not supplied. If an inherited ordinary calibration is treated as an outstanding theorem to be re-established, the final consistency conclusion is conditional on it; the new algebraic reductions remain independently proved. No assumption of Con(ZFC) is needed to establish an equivalence with Con(ZFC).

10.1 Offline replay

The release includes the literal polynomial packets, binary arithmetic DAGs, fixed coefficient payloads and finite reconstruction recipes, ordinary proof texts, source pins and notices, independent audits,

and this editable TeX. Python 3.10 or later and SymPy suffice for the complete arithmetic replay; no network access, theorem-prover installation, numerical solver, or blank machine simulation is required. From the extracted root run

```
python3 verify.py --output /tmp/con-zfc-replay.json.
```

The driver authenticates the exact file inventory, works in isolated copies, and invokes both normal and optimized Python checks explicitly. It also replays the selected original root-fusion and parity packages without changing them. The parity package deliberately rejects its entire driver under optimized Python; this expected policy result is recorded separately from its seven successful optimized algebraic auditors. It is not represented as an optimized whole-driver pass.

The assembled replay comprises 35 top-level commands in ten isolated groups. Seventeen literal artifacts are rebuilt byte-for-byte in each of two modes, giving 34 comparisons. Six archive-integrity corruption controls are additional to the mathematical mutation controls in the independent audits. These counts describe the packaged replay plan, not a count of proved theorems. The final ZIP receipt records the fresh-extraction result.

10.2 What changed and what remains

The new eleven-integer construction is a substantial combination of distinct ideas, not a claim that its individual ingredients originated here. The late intersective polynomial and full auxiliary interpolation theorem are credited; so are the earlier trace and parity methods. The centered budget removes the nearest-square coordinate, even strong ordinates make four radicands odd, and fixed parity weights replace all variable separation weights. The twelve- integer equation is the less expensive alternative. The fifteen-integer bounded-guard absorption and the positive one-sided residue bound are separate improvements.

The power-fusion interfaces eliminate special-function calls but still lack an ordinary-polynomial compilation competitive with these endpoints. The repository's direct C222 quartics and its nine-positive and ten-integer high-degree constructions remain relevant alternatives. No result here proves a lower bound or establishes a globally simplest equation.